

Hundred Days of Coding in C Programing

Day-7

Name: Arayana Khare

Batch:15

Sap id:590042432

Question 1

Write a C program to find acceleration when initial velocity, final velocity, and time are given.

1. Understand the problem statement and identify the required input, output, and formula.

Ans->

Given:

Initial velocity = u

Final velocity = v

Time = t

Acceleration = a

Formula:

$$a = \frac{v - u}{t}$$

Input:

u, v, t

Output:

Acceleration a

2. Prepare the standard test table.

Ans->

Test Case	Input(u,v,t)	Expected Output	Actual Output	Result
1	10,20,5	2	2	PASS
2	0,30,10	3	3	PASS
3	20,50,5	6	6	PASS

3. Write (no snippets) and evaluate the algorithm against the test cases prepared in Step 2.

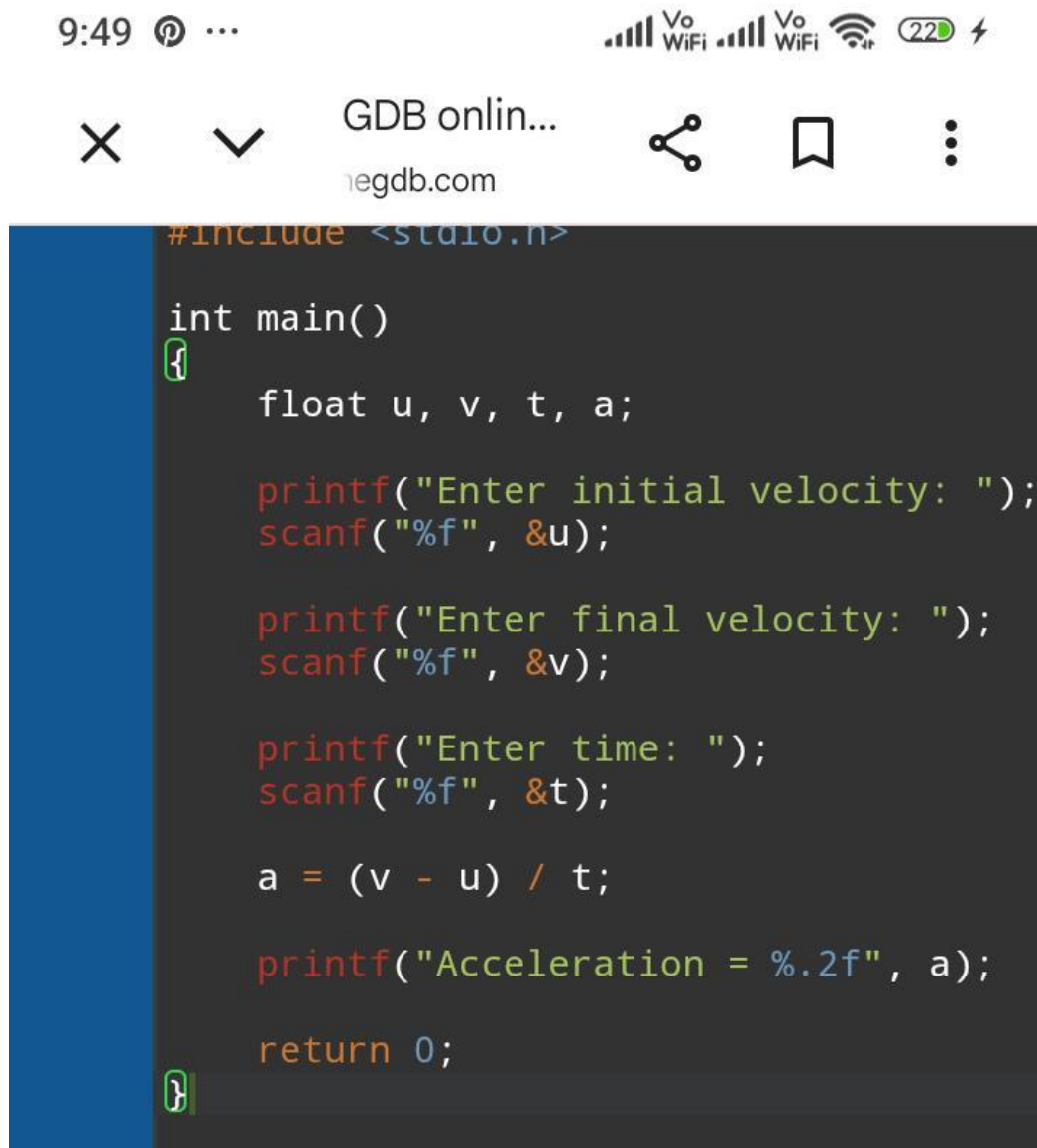
Ans->

Algorithm

1. Start
2. Declare variables u, v, t, a
3. Read initial velocity u
4. Read final velocity v
5. Read time t
6. Calculate acceleration using $a = (v-u)/t$
7. Display acceleration a
8. Stop

4. Write (no snippets) the C program.

Ans->



The screenshot shows a mobile browser interface. At the top, the status bar displays the time 9:49, a pin icon, and network status (VoWiFi, VoWiFi, and a Wi-Fi icon). Below the status bar, the browser address bar shows 'GDB online...' and 'regdb.com'. The main content area displays a C program for calculating acceleration. The code is as follows:

```
#include <stdio.h>

int main()
{
    float u, v, t, a;

    printf("Enter initial velocity: ");
    scanf("%f", &u);

    printf("Enter final velocity: ");
    scanf("%f", &v);

    printf("Enter time: ");
    scanf("%f", &t);

    a = (v - u) / t;

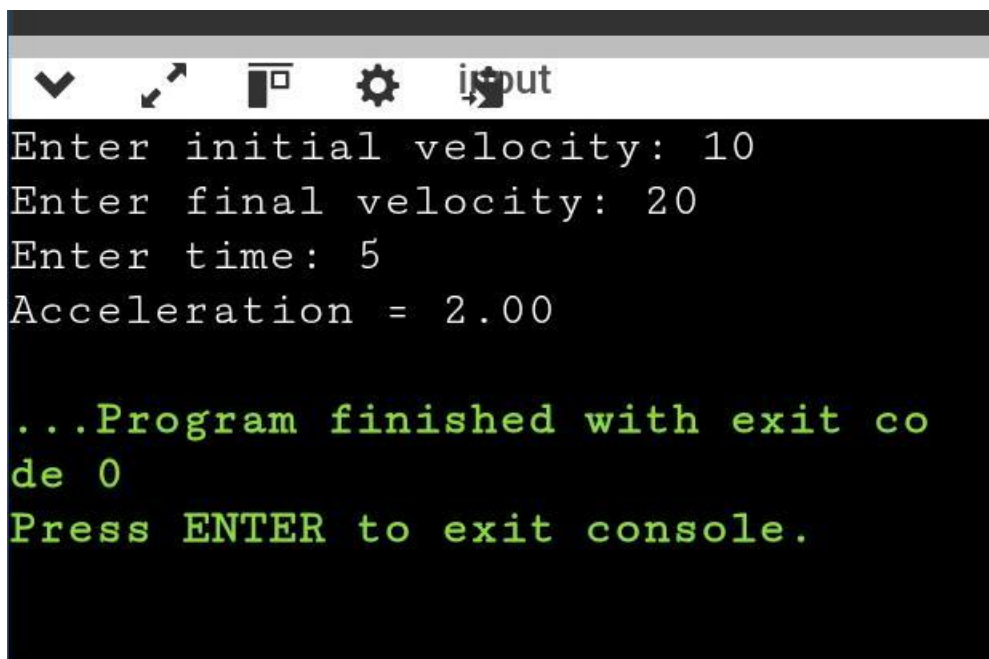
    printf("Acceleration = %.2f", a);

    return 0;
}
```

5. Compile and execute the program, then evaluate the code against the test cases prepared in Step 2.

Ans->

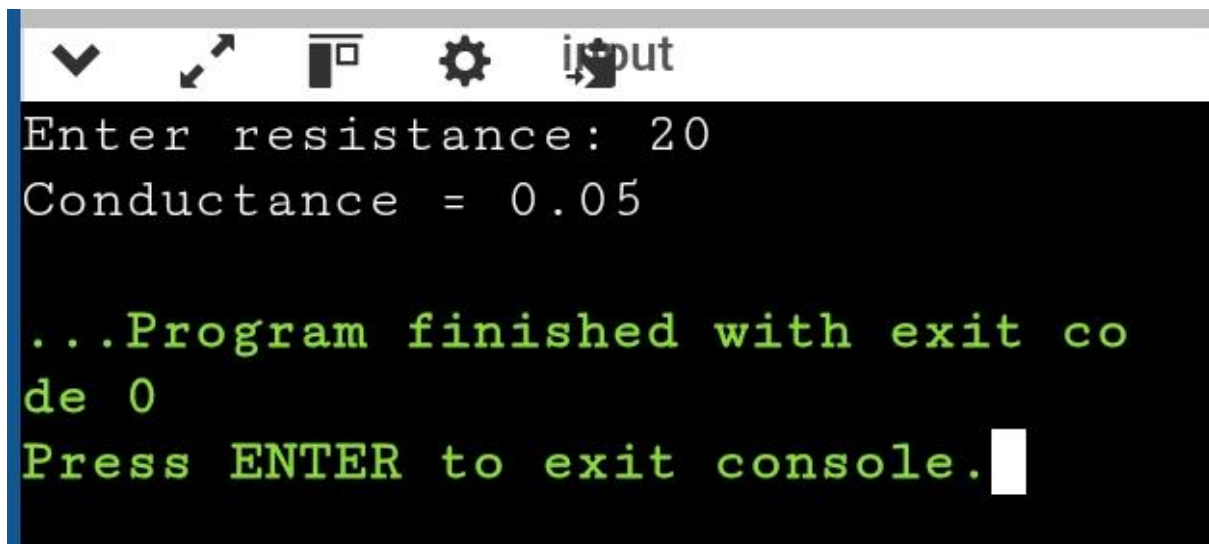
CASE 1:

A screenshot of a console window with a dark background and a light gray title bar. The title bar contains several icons: a downward arrow, a cursor, a square, a gear, and the word 'input'. The console text is as follows:

```
Enter initial velocity: 10
Enter final velocity: 20
Enter time: 5
Acceleration = 2.00

...Program finished with exit co
de 0
Press ENTER to exit console.
```

CASE 2:

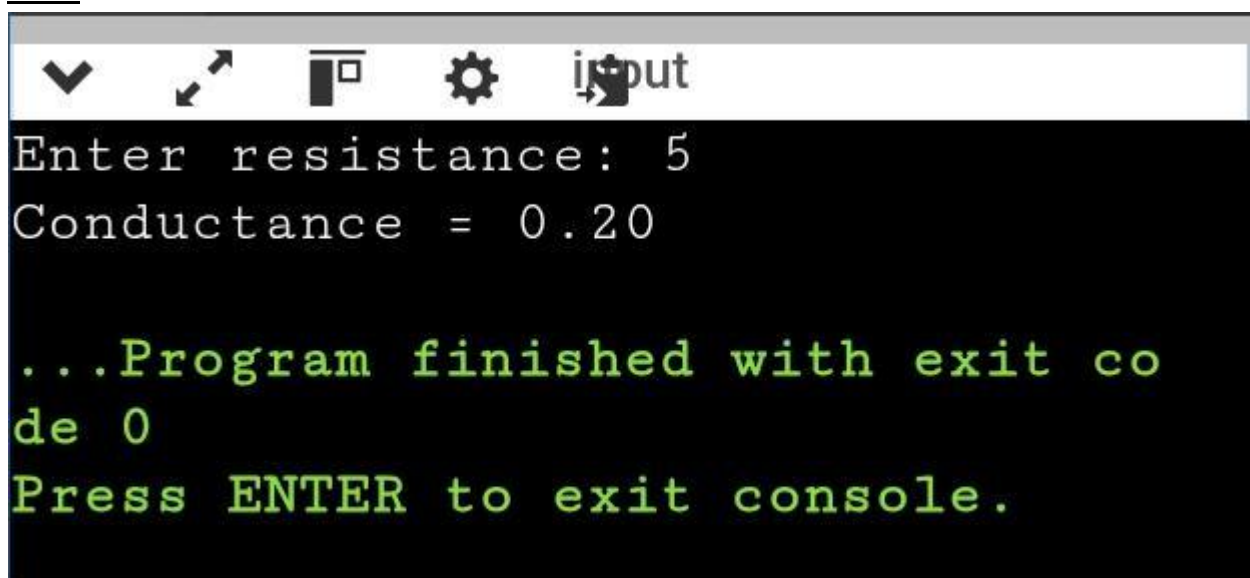


A terminal window with a title bar containing icons for window control and a label 'input'. The terminal text is as follows:

```
Enter resistance: 20
Conductance = 0.05

...Program finished with exit co
de 0
Press ENTER to exit console.
```

CASE 3:



A terminal window with a title bar containing icons for window control and a label 'input'. The terminal text is as follows:

```
Enter resistance: 5
Conductance = 0.20

...Program finished with exit co
de 0
Press ENTER to exit console.
```

6. Attach snippets/screenshots of compilation and execution.

Ans->

Compilation and Execution

CASE 1:

```
#include <stdio.h>

int main()
{
    float u, v, t, a;

    printf("Enter initial velocity: ");
    scanf("%f", &u);

    printf("Enter final velocity: ");
    scanf("%f", &v);

    printf("Enter time: ");
    scanf("%f", &t);

    a = (v - u) / t;

    printf("Acceleration = %.2f", a);

    return 0;
}
```

Enter initial velocity: 10
Enter final velocity: 20
Enter time: 5
Acceleration = 2.00

...Program finished with exit code 0
Press ENTER to exit console.

CASE 2:

```
#include <stdio.h>

int main()
{
    float u, v, t, a;

    printf("Enter initial velocity: ");
    scanf("%f", &u);

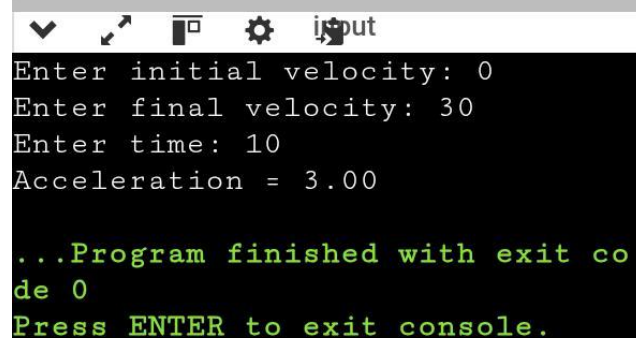
    printf("Enter final velocity: ");
    scanf("%f", &v);

    printf("Enter time: ");
    scanf("%f", &t);

    a = (v - u) / t;

    printf("Acceleration = %.2f", a);

    return 0;
}
```



Enter initial velocity: 0
Enter final velocity: 30
Enter time: 10
Acceleration = 3.00

...Program finished with exit code 0
Press ENTER to exit console.

CASE 3:


```
#include <stdio.h>

int main()
{
    float u, v, t, a;

    printf("Enter initial velocity: ");
    scanf("%f", &u);

    printf("Enter final velocity: ");
    scanf("%f", &v);

    printf("Enter time: ");
    scanf("%f", &t);

    a = (v - u) / t;

    printf("Acceleration = %.2f", a);

    return 0;
}
```

Enter initial velocity: 20
Enter final velocity: 50
Enter time: 5
Acceleration = 6.00

...Program finished with exit code 0
Press ENTER to exit console.

Question 2

Write a C program to find kinetic energy when mass and velocity are given.

1. Understand the problem statement and identify the required input, output, and formula.

Ans->

Given:

Mass = m

Velocity = v

Formula:

$$KE = \frac{1}{2}mv^2$$

Input:

m, v

Output:

Kinetic Energy KE

2. Prepare the standard test case table.

Ans->

Test Case	Input(m,v)	Expected Output	Actual Output	Result
1	10,4	80	80	PASS
2	5,6	90	90	PASS
3	20,10	1000	1000	PASS

3. Write (no snippets) and evaluate the algorithm against the test cases prepared in Step 2.

Ans->

Algorithm

- 1.Start
- 2.Declare variables m, v, KE
- 3.Read mass m
- 4.Read velocity v
- 5.Calculate $KE = 0.5 * m * v * v$
- 6.Display kinetic energy
- 7.Stop

4. Write (no snippets) the C program.

Ans->

```
#include <stdio.h>

int main()
{
    float m, v, KE;

    printf("Enter mass: ");
    scanf("%f", &m);

    printf("Enter velocity: ");
    scanf("%f", &v);

    KE = 0.5 * m * v * v;

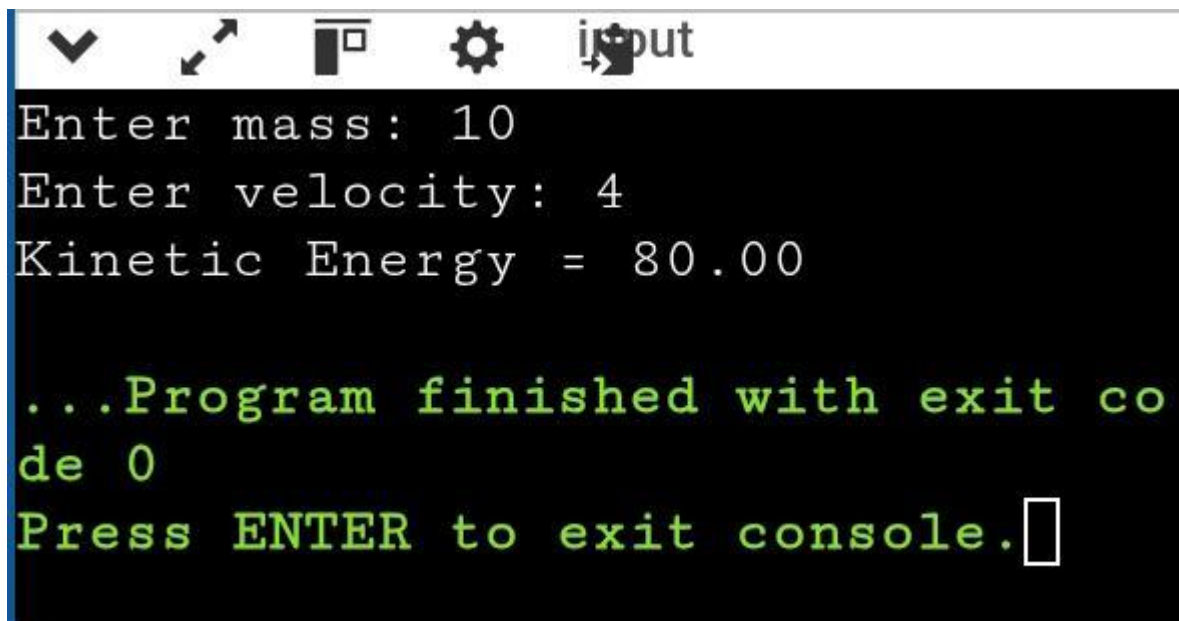
    printf("Kinetic Energy = %.2f", KE);

    return 0;
}
```

5. Compile and execute the program, then evaluate the code against the test cases prepared in Step 2.

Ans->

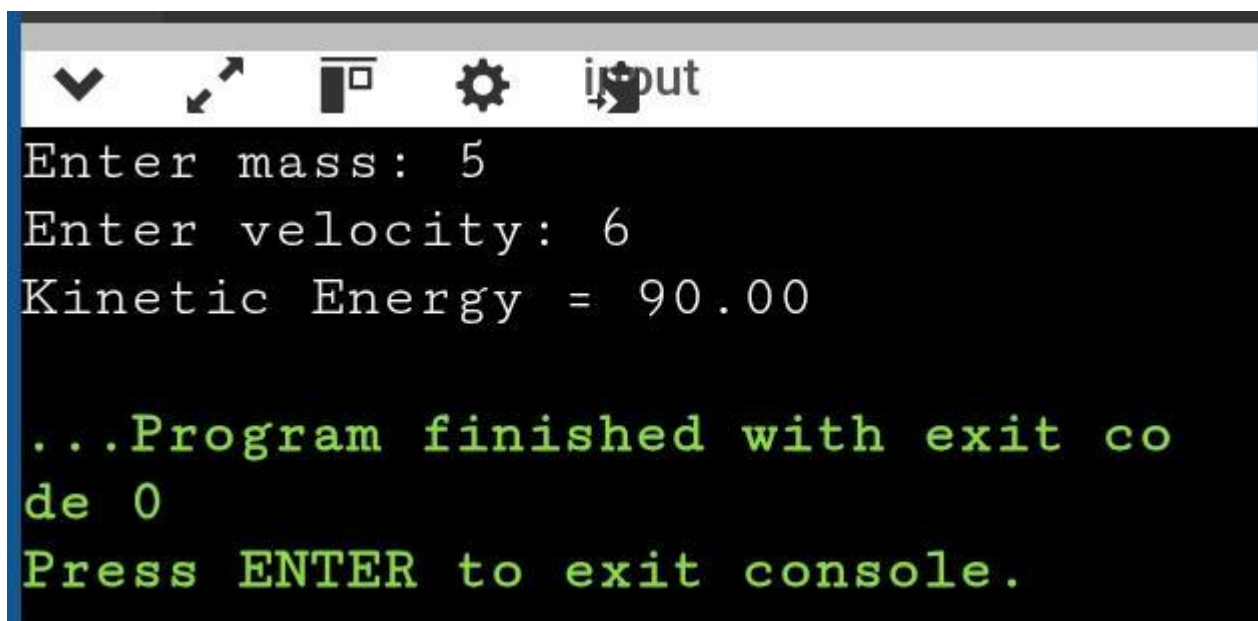
CASE 1:



```
Enter mass: 10
Enter velocity: 4
Kinetic Energy = 80.00

...Program finished with exit co
de 0
Press ENTER to exit console.
```

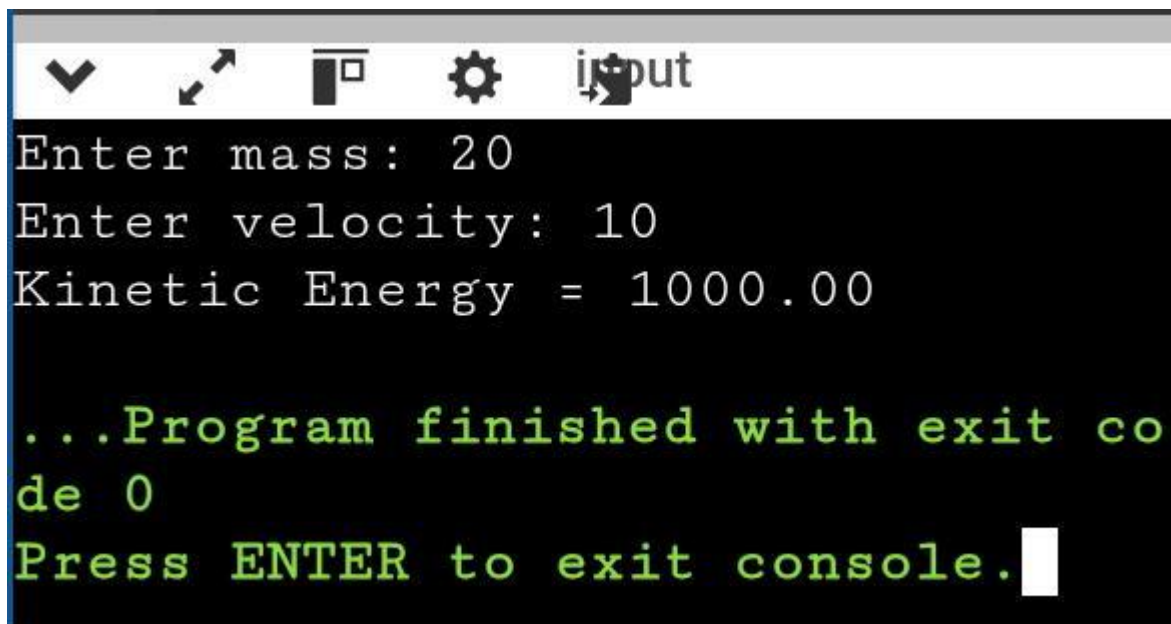
CASE 2:



```
Enter mass: 5
Enter velocity: 6
Kinetic Energy = 90.00

...Program finished with exit co
de 0
Press ENTER to exit console.
```

CASE 3:

A screenshot of a console window with a dark background. The title bar at the top is light gray and contains several icons: a downward arrow, a double-headed arrow, a square icon, a gear icon, and the word 'input'. The console text is as follows: 'Enter mass: 20' (white), 'Enter velocity: 10' (white), 'Kinetic Energy = 1000.00' (white), '...Program finished with exit co' (green), 'de 0' (green), and 'Press ENTER to exit console.' (green) followed by a white cursor block.

```
Enter mass: 20
Enter velocity: 10
Kinetic Energy = 1000.00

...Program finished with exit co
de 0
Press ENTER to exit console.
```

6. Attach snippets/screenshots of compilation and execution.

Ans->

Compilation and Execution

CASE 1:

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      float m, v, KE;
6
7      printf("Enter mass: ");
8      scanf("%f", &m);
9
10     printf("Enter velocity: ");
11     scanf("%f", &v);
12
13     KE = 0.5 * m * v * v;
14
15     printf("Kinetic Energy = %.2f",
16
17     return 0;
18 }
```

Enter mass: 20
Enter velocity: 10
Kinetic Energy = 1000.00

...Program finished with exit code 0
Press ENTER to exit console.

CASE2:

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      float m, v, KE;
6
7      printf("Enter mass: ");
8      scanf("%f", &m);
9
10     printf("Enter velocity: ");
11     scanf("%f", &v);
12
13     KE = 0.5 * m * v * v;
14
15     printf("Kinetic Energy = %.2f", KE);
16
17     return 0;
18 }
```

Enter mass: 10
Enter velocity: 4
Kinetic Energy = 80.00

...Program finished with exit code 0
Press ENTER to exit console.

CASE 3:

```

1  #include <stdio.h>
2
3  int main()
4  {
5      float m, v, KE;
6
7      printf("Enter mass: ");
8      scanf("%f", &m);
9
10     printf("Enter velocity: ");
11     scanf("%f", &v);
12
13     KE = 0.5 * m * v * v;
14
15     printf("Kinetic Energy = %.2f",
16
17     return 0;
18 }

```



```

Enter mass: 5
Enter velocity: 6
Kinetic Energy = 90.00

...Program finished with exit co
de 0
Press ENTER to exit console.

```

Question 3

Write a C program to find conductance when resistance is given.

1. Understand the problem statement and identify the required input, output, and formula.

Ans->

Given:

Resistance = R

Formula: $G=1/R$

Where:

G = Conductance

R = Resistance

Input:

Resistance R

Output:

Conductance G

2. Prepare the standard test case table.

Ans->

Test Case	Input	Expected Output	Actual Output	Result
1	10	0.1	0.1	PASS
2	20	0.05	0.05	PASS
3	5	0.2	0.2	PASS

3. Write (no snippets) and evaluate the algorithm against the test cases prepared in Step 2.

Ans->

Algorithm

1. Start
2. Declare variables R and G
3. Read resistance R
4. Calculate $G = 1/R$
5. Display conductance G
6. Stop

4. Write (no snippets) the C program.

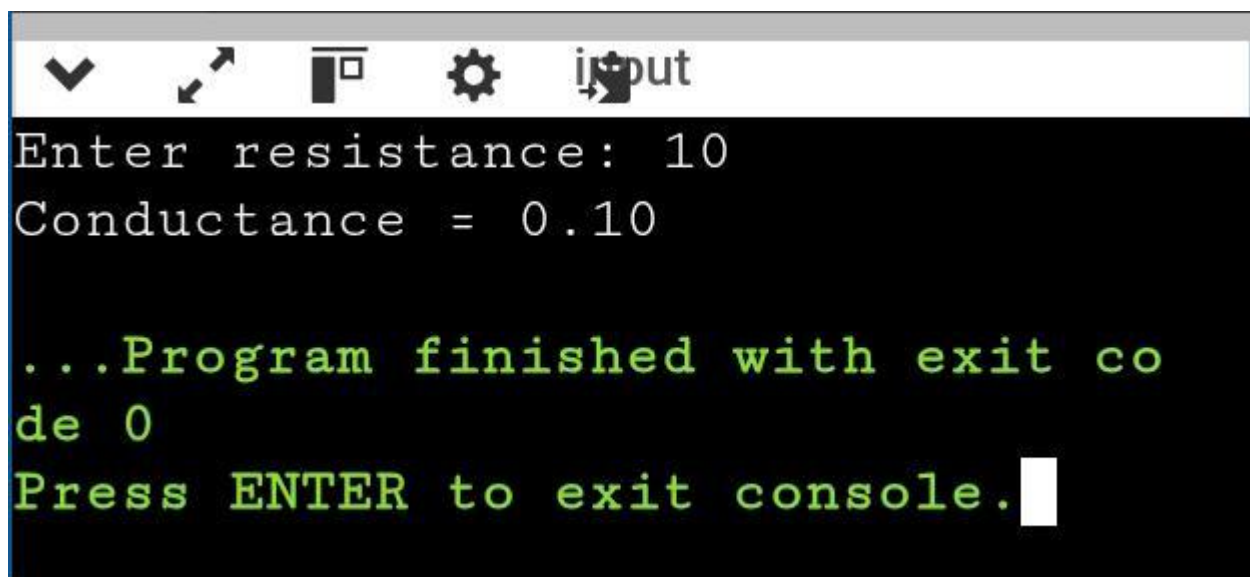
Ans->

```
source code
1  #include <stdio.h>
2
3  int main()
4  {
5      float R, G;
6
7      printf("Enter resistance: ");
8      scanf("%f", &R);
9
10     G = 1 / R;
11
12     printf("Conductance = %.2f", G);
13
14     return 0;
15 }
```

5. Compile and execute the program, then evaluate the code against the test cases prepared in Step 2.

Ans->

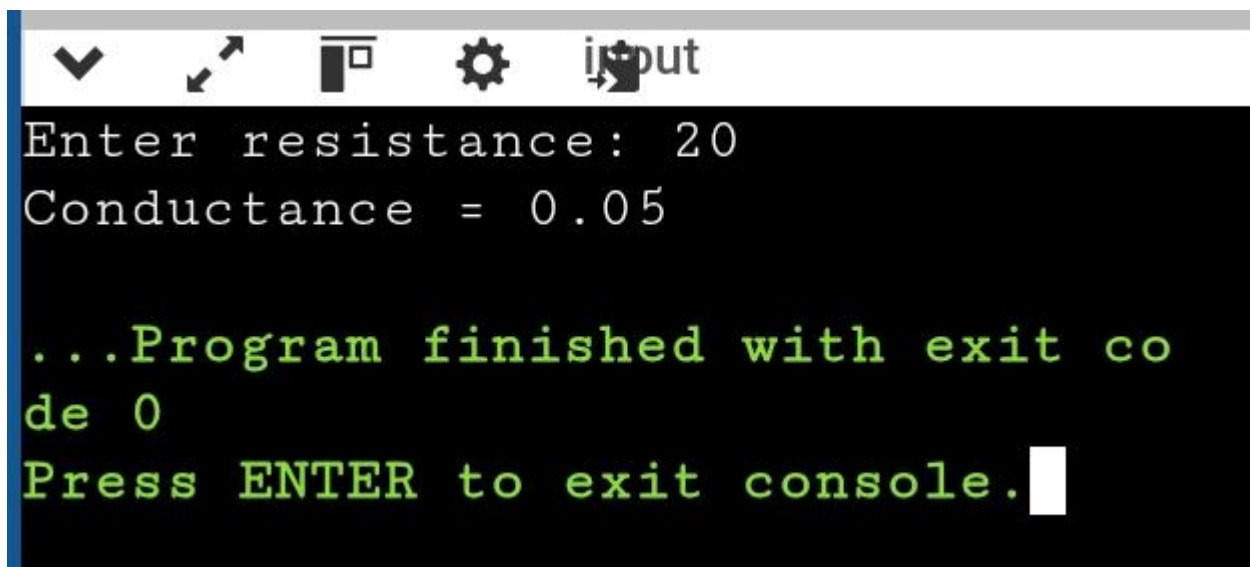
CASE 1:



```
Enter resistance: 10
Conductance = 0.10

...Program finished with exit co
de 0
Press ENTER to exit console.
```

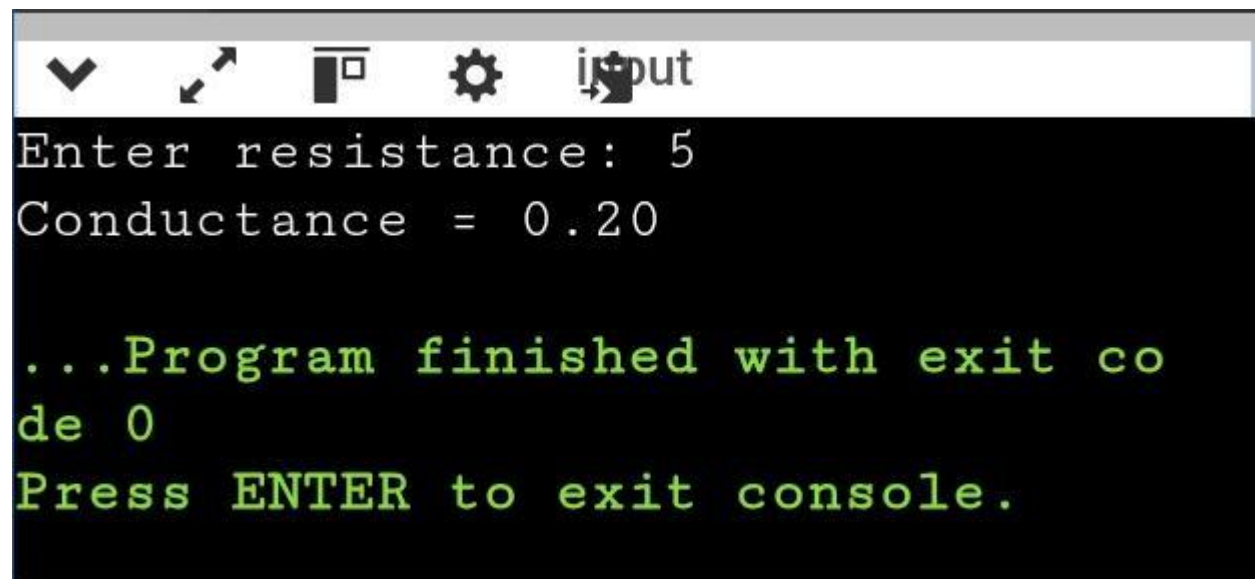
CASE 2:



```
Enter resistance: 20
Conductance = 0.05

...Program finished with exit co
de 0
Press ENTER to exit console.
```

CASE 3:

A screenshot of a terminal window. The title bar is light gray and contains several icons: a downward arrow, a window with an arrow pointing out, a square icon, a gear icon, and the word "input" with a mouse cursor icon. The terminal background is black, and the text is white. The text displayed is: "Enter resistance: 5", "Conductance = 0.20", "...Program finished with exit co", "de 0", and "Press ENTER to exit console." in a green monospace font. A thick horizontal bar is visible below the terminal window.

```
Enter resistance: 5
Conductance = 0.20

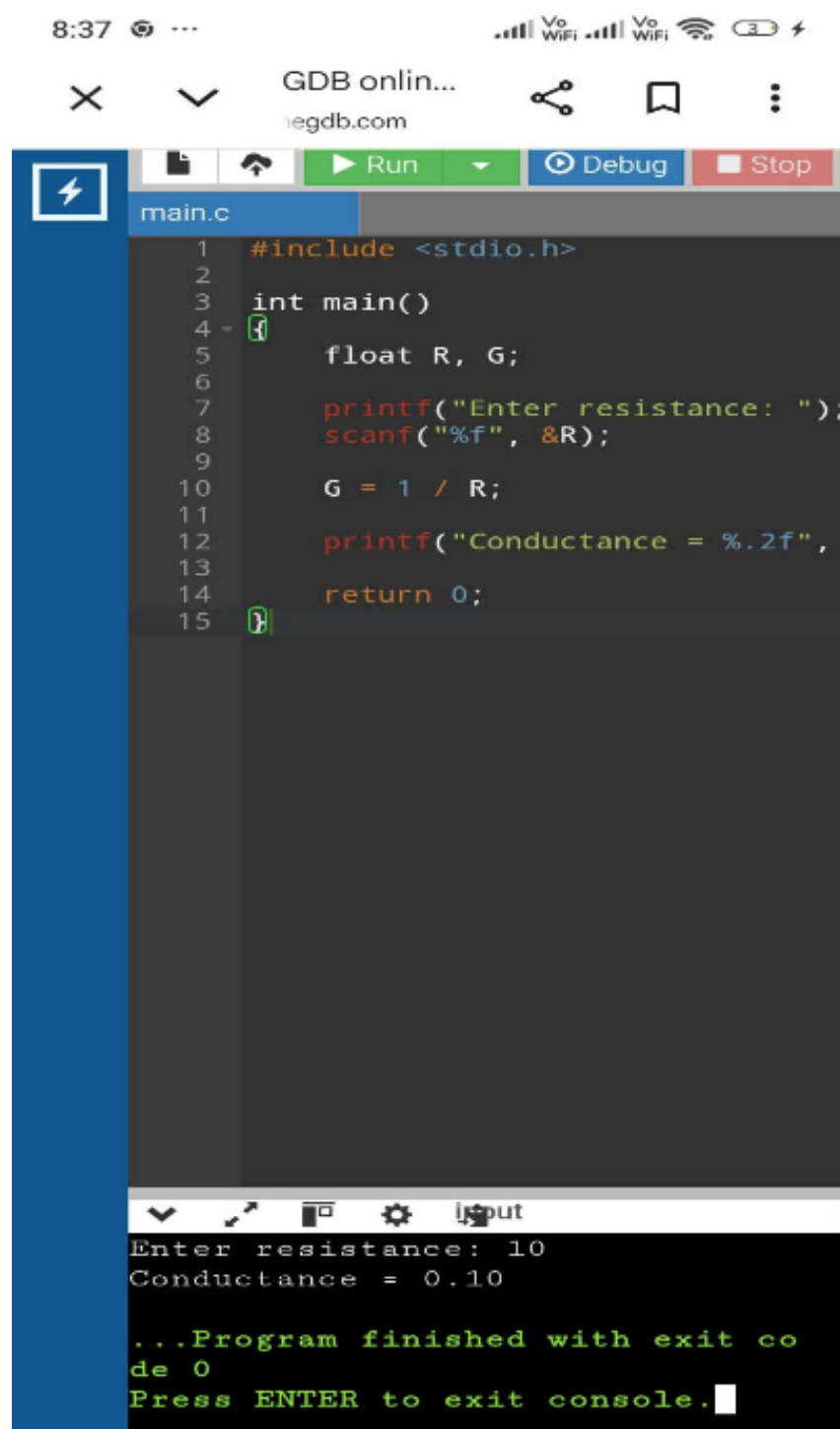
...Program finished with exit co
de 0
Press ENTER to exit console.
```

6. Attach snippets/screenshots of compilation and execution.

Ans->

Compilation and Execution

CASE 1:



8:37 8:37 ... Vo WiFi Vo WiFi 3 4

GDB online...
egdb.com

main.c

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float R, G;
6
7     printf("Enter resistance: ");
8     scanf("%f", &R);
9
10    G = 1 / R;
11
12    printf("Conductance = %.2f",
13
14    return 0;
15 }
```

Enter resistance: 10
Conductance = 0.10

...Program finished with exit code 0
Press ENTER to exit console.

CASE 2:

8:37 Vo WiFi Vo WiFi 3

GDB online... regdb.com

Run Debug Stop

main.c

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float R, G;
6
7     printf("Enter resistance: ");
8     scanf("%f", &R);
9
10    G = 1 / R;
11
12    printf("Conductance = %.2f",
13
14    return 0;
15 }
```

input

Enter resistance: 20
Conductance = 0.05

...Program finished with exit code 0
Press ENTER to exit console.

CASE 3:

8:37 ... Vo WiFi Vo WiFi 3

GDB online... regdb.com

Run Debug Stop

main.c

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float R, G;
6
7     printf("Enter resistance: ");
8     scanf("%f", &R);
9
10    G = 1 / R;
11
12    printf("Conductance = %.2f",
13
14    return 0;
15 }
```

Enter resistance: 5
Conductance = 0.20

...Program finished with exit code 0
Press ENTER to exit console.

